

What Networking Got Right That Agentic AI Risks Getting Wrong

The Case for an Agent Control Plane

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INTEROPERABILITY AND STANDARDS TRACK
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The Nokia logo is displayed in a dark blue, sans-serif font. It is centered within a large, light blue circle that has a white border. The background of the slide is a dark blue gradient.

The internet's answer to interoperability: the autonomous system (AS)

When the internet evolved to a global scale, Networks needed a unit smaller than the whole network and bigger than one router unit of work to scale : **an autonomous, boundary-declaring domain.**

01



Each domain kept its own internal architecture

the Autonomous System — AS

No two networks had to be built the same way.

02



Domains declared only what crossed their boundary

route advertisement / peering policy

Never their internal policy.

03

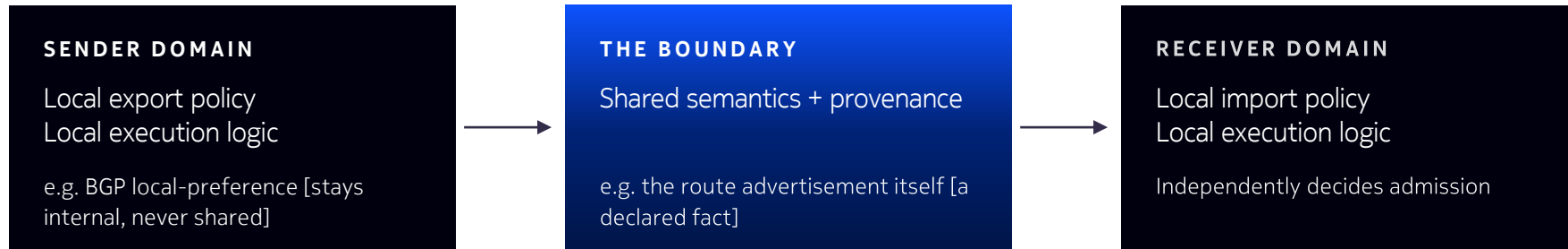


Every other domain could rely on that declaration

AS path / accepted routes

Trust without inspection.

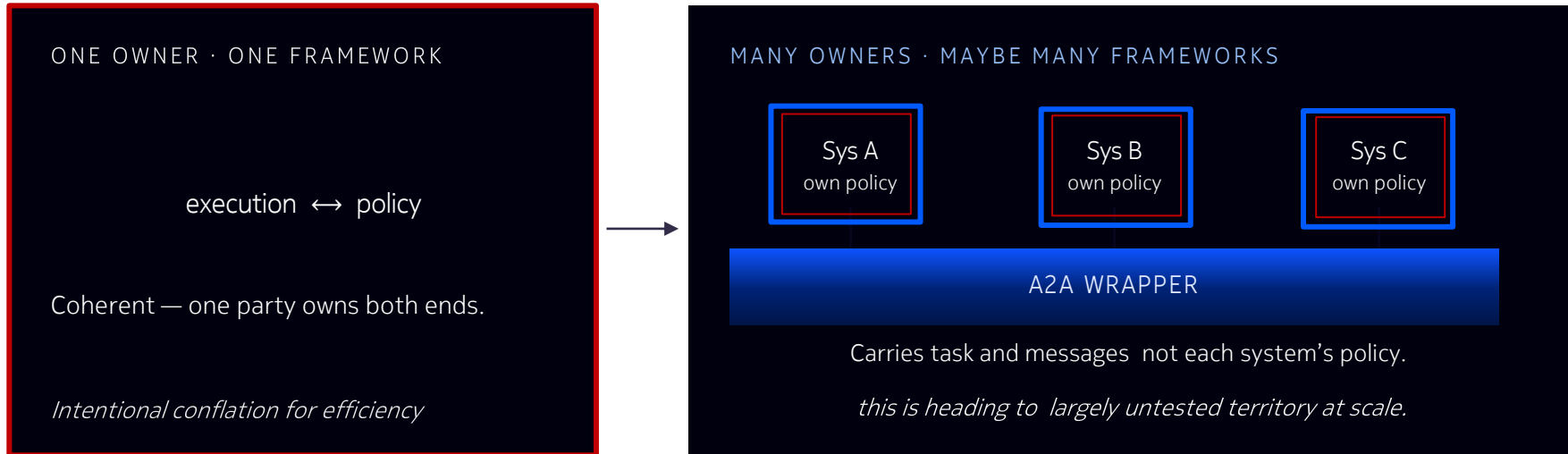
Networking standardized what policy could act on not the policy itself.



BGP never asked every network to agree on why they prefer a route — only on what a route advertisement means.

Standardize the semantics of the crossing, not the policy of the domain.

Works within one system. Untested across many.



Inter-domain agentic chains require separation of execution logic from policies.

Policies need to be translated and retained at the right level to prevent silent failures and over-permission.

Federating Agents Is Not Enough. Federate Domains.

An **Agent Control Domain (ACD)**, as a first-class unit is the **largest entity that can make and enforce coherent assertions about authority, behavior/delegation and data.**

BGP · AUTONOMOUS SYSTEM

Not a router.

An administrative and control boundary containing routers.

Declares what it will and will not transit.

The unit of inter-domain reasoning.

AGENT CONTROL DOMAIN (ACD)

Not an agent.

An administrative and control boundary within which agents operate.

Declares permitted crossing classes.

The unit of cross-domain agent reasoning.

INVARIANT 1

The ACD is the unit of inter-domain control not the agent.

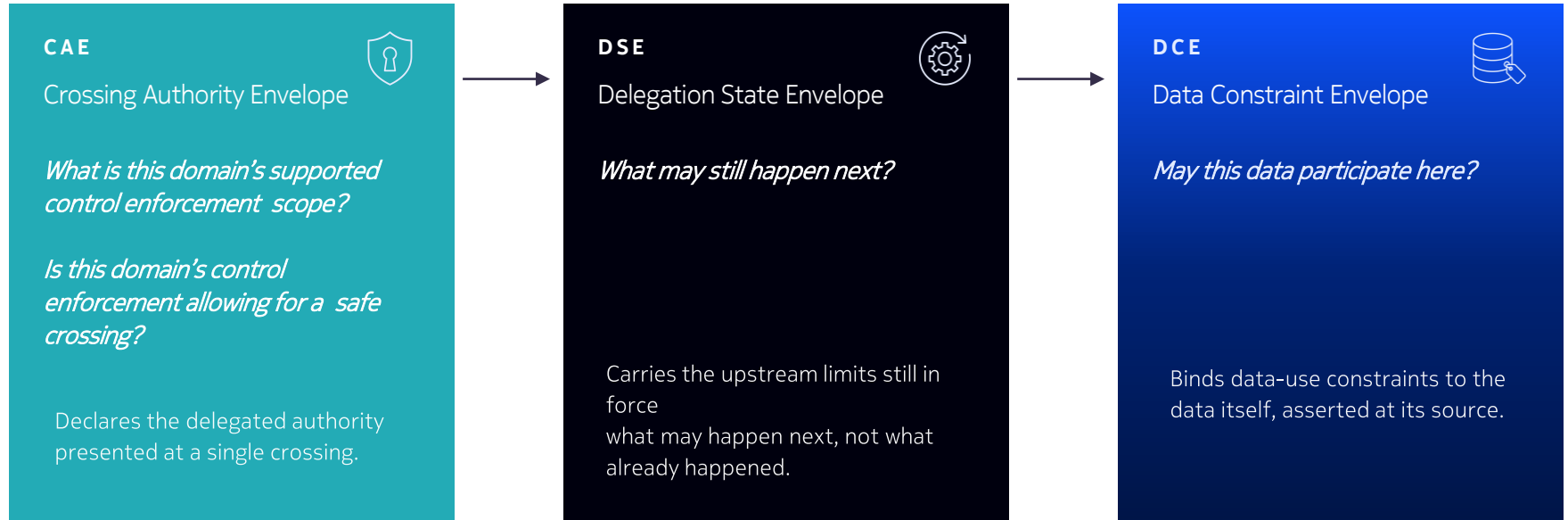
INVARIANT 2

The boundary requires modeling of crossing classes; the crossing classes will act as permission units as well as control enforcement capabilities that could be advertised.

BEYOND A TOKEN · keyed by domain, the crossing classes are few enough and slow enough to publish and converge. Keyed by agent or task, nothing ever converges.

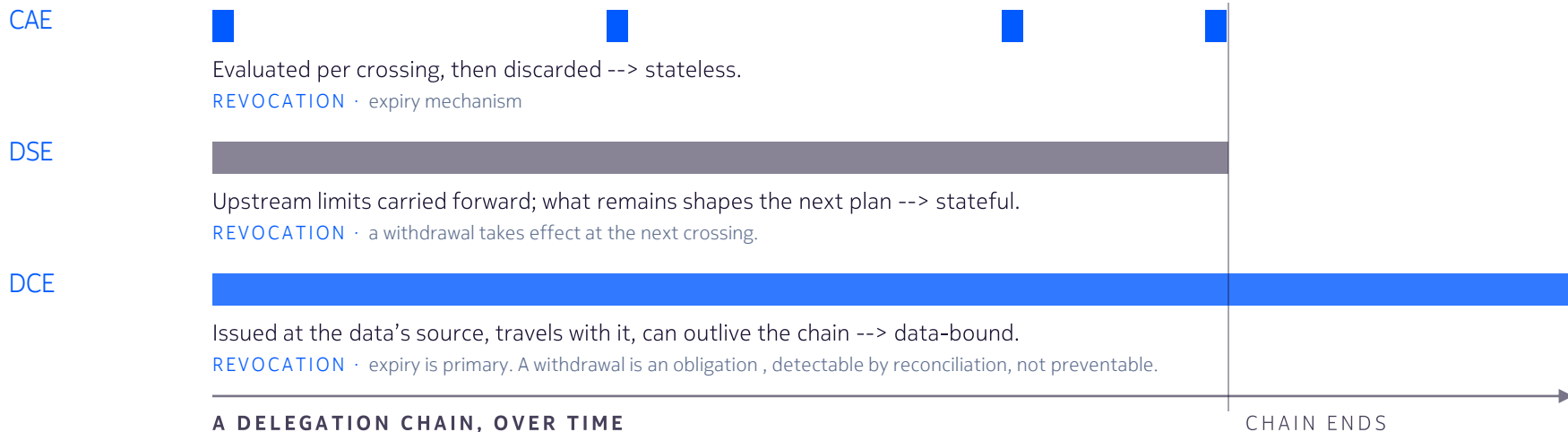
One request, three declarations

If a domain's interior is opaque, something still has to be declared at its boundary — **on behalf of authority, delegation state, and data.**



WHY ONE TOKEN CAN'T DO THIS

Not three fields. Three different lifetimes.



CAE allow, DSE allow, DCE deny is coherent — B may call C, the delegation is valid, A's data still may not travel. One token cannot say that, because the three don't live on the same clock.

INVARIANT 3

A domain cannot self-grant scope it wasn't delegated



PRE-AUTHORIZED SCOPE ACTIVATION

A downstream domain never expands its own authority.

An upstream authority may pre-grant **a specific, time-bounded, named scope** that activates only under defined conditions — always traceable to an authority that actually held it.

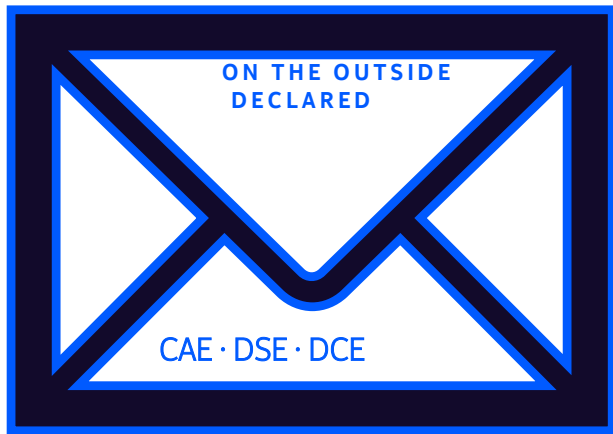
Activation of existing authority, not amplification.

INVARIANT 3

Authority only narrows; it never inflates in downstream trajectory of a chain over domains.

INVARIANT 4 and 5

Enforcement lives at the boundary, outside the agent's logic. That's what makes it interoperable.



Decides on declared artifacts

CAE, DSE, DCE — plus the receiving domain's own local policy.

Never inspects payload or internal reasoning

Control stays independent of the agent's internals.

Says nothing about how the work gets done

Interior reasoning is out of scope. Onward crossings are not — that is the DSE.

INVARIANT 4

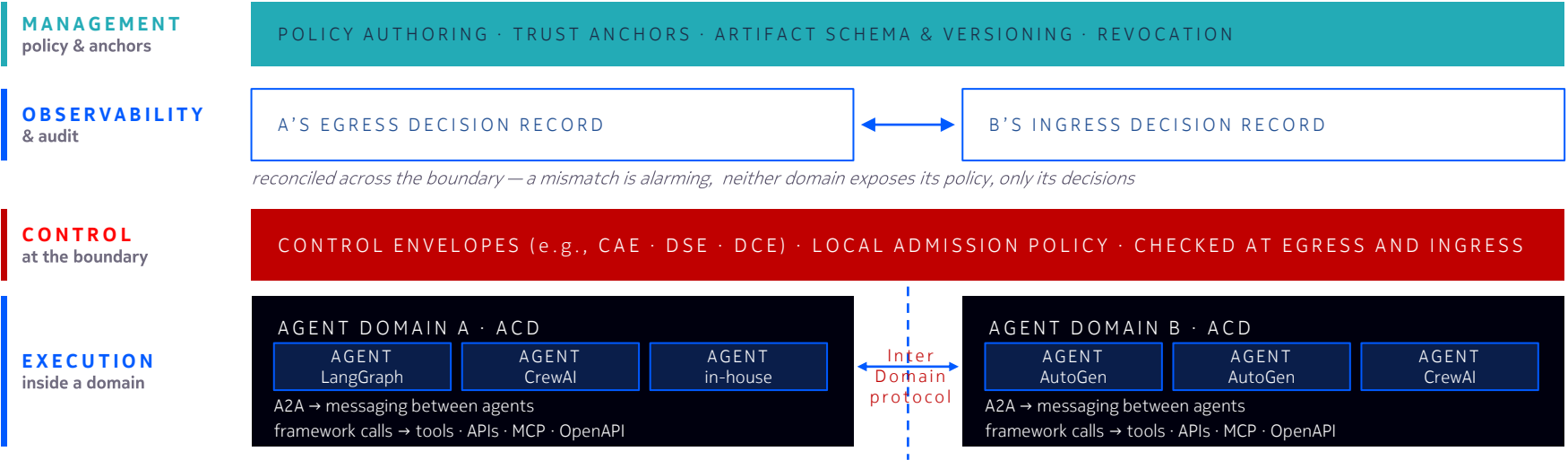
A chain is only as good as its anchor. Self-asserted authority, with no anchor the receiver recognizes, is not authority.

INVARIANT 5

The execution semantics and agent internal logics are opaque to the control plane.

WHERE EACH PLANE LIVES

Execution inside. Control at the boundary.



Control Envelopes and Networking IP/MPLS headers are similar in operation

A DOMAIN HEADER WRAPS ONE OR MORE AGENT MESSAGES · THE PEER EDGE READS ONLY THE HEADER

THE INTERNET · IP, MPLS, GRE

IP / MPLS HEADER · READ AT EVERY BOUNDARY

Imposed at ingress, popped at egress. Addressed to a network, never to a process inside it.

PAYLOAD · OPAQUE IN TRANSIT

TCP segments and application data, forwarded without being read, understood or rewritten by anyone in between.

THE AGENT STACK · ACDs AND CROSSINGS

AGENT DOMAIN HEADER · READ AT BOTH EDGES

CAE — class, subject, purpose, limits · the DSE chain so far · an DCE digest per data object.

PAYLOAD · OPAQUE TO THE CONTROL PLANE

One or more A2A or MCP messages, carried unmodified [never inspected, never re-signed, never verified by Domain boundary]

HOW THE ANALOGY MAPS · THREE PROPERTIES CARRIED OVER

ADDRESSED TO A DOMAIN, NOT A HOST

The domain header characterizes what the receiving ACD is expected to enforce

IMPOSED ONCE, HONOURED FOR A WHILE

A crossing context authorizes a class once, and bounds the exchange with limits.

OPACITY BY CONSTRUCTION

The core forwards what it cannot read; the peer executes what we do not inspect.

THE PAYOFF · NO PROTOCOL COMPETITION OR DIRECT DEPENDENCY

Two layers, one boundary — and no negotiation about what happens on the other side.

THE MISSING COMPOSITION POINT

The pieces exist. The boundary does not.

AUTHORIZATION, PROVENANCE AND DATA POLICY PRODUCE DECLARATIONS · THE ACD MAKES THEM ONE DECISION

FOUR THINGS THE ACD OFFERS THAT NO EXISTING MECHANISM DOES

01 Abstraction of the interior

Internal implementations and local policy stay private. Only the crossing is exposed.

what A2A's task modeling left with nowhere to live.

02 A declarative surface

A domain declares the classes of work it admits and the obligations it can enforce.

published and slow-moving, not renegotiated per peer.

03 Run-time control of a Chain across domains

The same declarations govern one request chain across independently run systems.

asserted at every boundary, not only at design time.

04 One unit to bind to

Data constraints, delegation state and authority attach to one named scope.

previously each bound elsewhere, or to nothing at all.

THE ACD IS THE MISSING LOCUS OF CONTROL · IT RECEIVES THE DECLARATIONS, APPLIES PRIVATE LOCAL POLICY, AND ACCEPTS OR REFUSES THE CROSSING

The landscape solves connection, not crossing.

TOOL-INVOCATION



MCP · OpenAPI tool calling

Tool frameworks expose capabilities. Standardize how an agent reaches a tool.

STEWARDS · Linux Foundation (MCP) · OpenAPI Initiative

AGENT-TO-AGENT



A2A · ACP · ANP

Carry the request between agents, tasks and artifacts. Don't carry the control semantics of the crossing.

STEWARDS · Linux Foundation (A2A, ACP) · ANP community project

IDENTITY & TRUST



OAuth 2.x · OIDC · SPIFFE · VCs

Authorization establishes scoped authority. Establish who is calling.

STEWARDS · IETF · OpenID Foundation · CNCF · W3C

Each family answers a real question, but none focus on a domain as the foundational data model and crossings.

A semantics model first. The protocol question comes after.

WHAT WAS PROPOSED



- The **domain (ACD)** as a boundary primitive
- Three distinct control concerns: **authority, cumulative behavior, data-use**
- A small set of **cross-domain invariants** ; 5 to start with
- **Protocol- and topology-neutral**

OPEN QUESTIONS · INITIAL MATERIAL IN BACKUP



- Wire format and signing of the three envelopes
- minimum crossing-class vocabulary,
- lifecycle and revocation,
- binding constraints to requests, delegation state, and data.
- and many more

Networking scaled because independently built domains learned a common boundary vocabulary and procedures without surrendering control of their interiors.

Agent ecosystems need the same kind of agreement and their own tailored protocols.



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AN INVITATION TO COLLABORATION:

What is the minimum control semantics every agent domain should understand at its boundary?

NOKIA

Backup slides

IS THIS JUST ANOTHER A2A EXTENSION

Part of it is. The part that matters breaks the assumptions in A2A.

WHAT IF IT RIDES ON A2A TODAY:

DOMAIN IN THE AGENT CARD · CAE ON MESSAGE SEND · DSE BOUND TO PARTS AND ARTIFACTS

WHERE THE ASSUMPTIONS BREAK

01 · PAIRWISE BY CONSTRUCTION

A client and a remote agent. A cumulative chain artifact has no owner in a two-party model, and no one is obliged to carry it forward.

02 · NOT EVERY HOP SPEAKS A2A

Chains run through tool calls, queues, batch jobs and plain APIs. Control that exists only inside A2A stops at the first hop that is not A2A.

03 · THE DATA OUTLIVES THE SESSION

The task completes, the connection closes, the data persists. A transport-scoped extension cannot bind data at rest or govern its reuse tomorrow.

04 · EXTENSIONS ARE NEGOTIATED

An optional, per-peer capability that can be declined is not control. Admission has to be the receiving domain's decision, not a handshake.

THE VERDICT · A GOOD FIRST CARRIER · NOT THE ONLY ONE AND NOT THE BEST ONE · THE ARTIFACTS MUST BE TRANSPORT-INDEPENDENT

Define the semantics once independent of A2A later decide whether we should bind them to A2A.

One ask: Agree the primitives first.

THE CLAIM · EACH LAYER PERMITS ONLY WHAT ITS OWN UNIT CAN NAME · THE CROSSING IS THE UNIT NO LAYER HAS

THE UNIT EACH LAYER WORKS ON · AND WHAT IT CAN THEREFORE DECIDE

TOKENS · MODELS
GATEWAYS, ROUTERS
How much may be consumed

THE TOOL CALL
MCP
Which capability may be invoked

THE AGENT
A2A · IDENTITY AND TRUST
Who acts, for whom, with what approval

THE CROSSING
THIS WORK
Whether work may leave one domain and enter another

THE FOUR QUESTIONS THIS ROOM ASKS

01 · IS THIS ALREADY BEING DONE?
A DIFFERENT UNIT ENTIRELY

None of the layers below can name a boundary, so none of them can permit a crossing.

02 · WHOSE WORKING GROUP IS THIS?
SAME TERMS, DIFFERENT MEANING
Move delegation, trust and audit to the crossing and their semantics, interactions and operations all change.

03 · SO YOU ARE REPLACING A2A?
IT IS CARRIED, NOT REPLACED
A2A crosses unparsed and stays in action. It sees one connection; what follows inside the domain it never sees.

04 · ISN'T THE GATEWAY ALREADY THIS?
DIFFERENT DOORS, DIFFERENT PRIMITIVES
Each gateway sits at its own entry or exit with its own primitives. None of them sits at the domain boundary.

THE CALL · **AGREE THE PRIMITIVES, THEIR SEMANTICS, ACTORS AND OPERATIONS** · THEN DECIDE TOGETHER WHERE THAT WORK BELONGS

This may end as a protocol, or as extensions to several — [what is not in question is that the primitives have to be agreed before either.](#)

APIs bound the call. Nothing bounds the consequence.

CONTRACTS AND MESSAGE FORMATS ARE NECESSARY BUT NEITHER IS A BOUNDARY

What ACD offers as a first-class unit

01

REFUSE BY CLASS

Decline a kind of work from a domain — before knowing the agent.

APIs: identify the caller first.

02

CONSTRAINTS THAT TRAVEL

Obligations ride with the data, still binding when B calls C.

APIs: no contract reaches hop two.

03

ONE SHARED RECORD

Release and acceptance reconcile without exposing policy.

APIs: logs are one-sided.

04

POLICY OUTLIVES THE STACK

The class survives a change of vendor, framework or model.

APIs: rewrite and renegotiate.

WHERE APIs ARE ENOUGH · ONE OWNER · FIXED INTEGRATIONS · DETERMINISTIC CLIENTS

It earns its place when **counterparties are many, topology shifts without renegotiation, and the caller decides what to call.**

FUNCTIONAL VIEW · WHAT THE PROTOCOL ACTUALLY DOES

Four functions. Two sides. One shared record.

IDENTITY WHO IS A DOMAIN	DOMAIN IDENTITY Named, not per-agent.	TRUST ANCHORS Who may assert a class.	MEMBERSHIP Agents map to a domain.
POLICY WHAT MAY CROSS	CROSSING CLASSES Kinds of work, published.	ACCEPT AND REFUSE Stated before identity.	NO INTERNALS Policy stays private.
CROSSING THE ACT ITSELF	EGRESS DECISION Release, with obligations.	ENVELOPES CAE · DSE · DCE over an opaque payload.	INGRESS DECISION Accept, refuse or constrain.
EVIDENCE WHAT REMAINS	CANONICAL FORM One byte-exact rendering.	TWO SIGNATURES Release and acceptance.	RECONCILE Both sides, same record.
IN-DOMAIN EXECUTION STAYS OPAQUE · A2A AND MCP RIDE INSIDE, UNPARSED			

The control plane is slow and agreed. The crossing is fast and local. Only the record is shared.